Hemoglobinopathies & Thalasemia

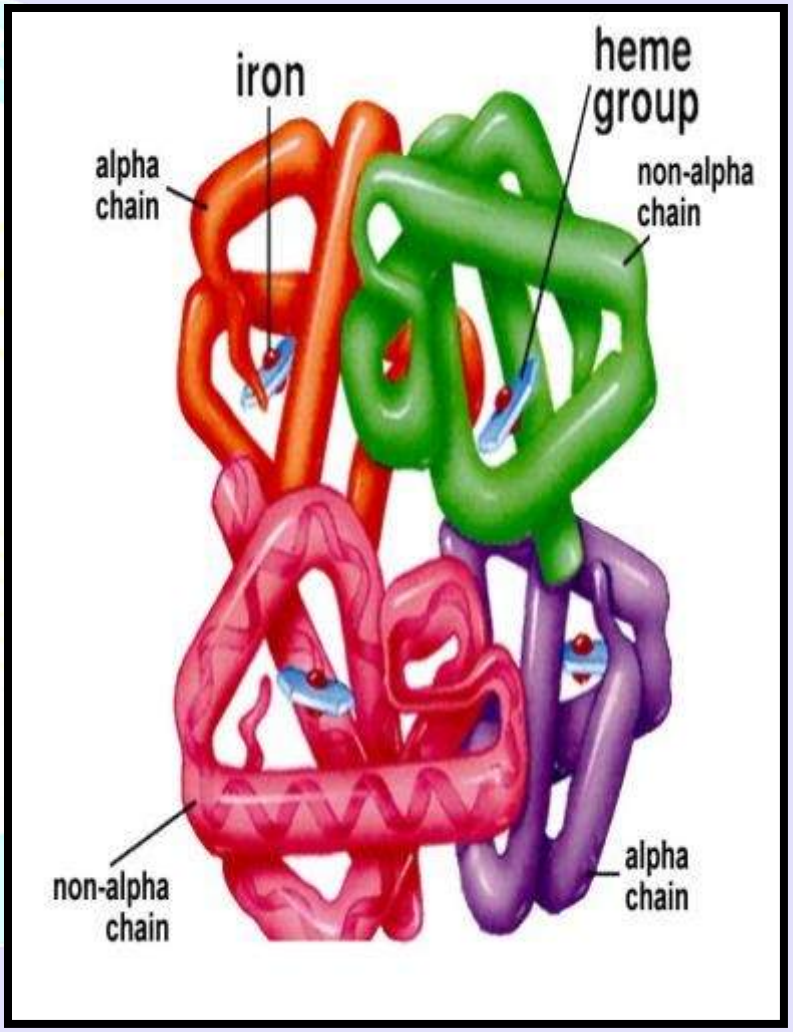
LEARNING OBJECTIVES

At the end of the lecture student should be able to;

- **Define hemoglobinopathies**
- **Appreciate normal globin chain configuration**
- **Define and classify thalassemias**
- **Appreciate four possible alpha thalassemia syndromes**
- **Explain type of mutations responsible for beta thalassemia**
- **Lab test responsible for diagnosing hemoglobinopathy and thalassemias**

HEMOGLOBINOPATHIES

- **Hemoglobinopathy is a kind of genetic defect that results in abnormal structure of one of the globin chains of the hemoglobin molecule**
- **Hemoglobinopathies are inherited single-gene disorders; in most cases, they are inherited as autosomal co-dominant traits**
- **It is estimated that 7% of worlds population (420 million) are carriers, with 60% of total and 70% pathological being in Africa**

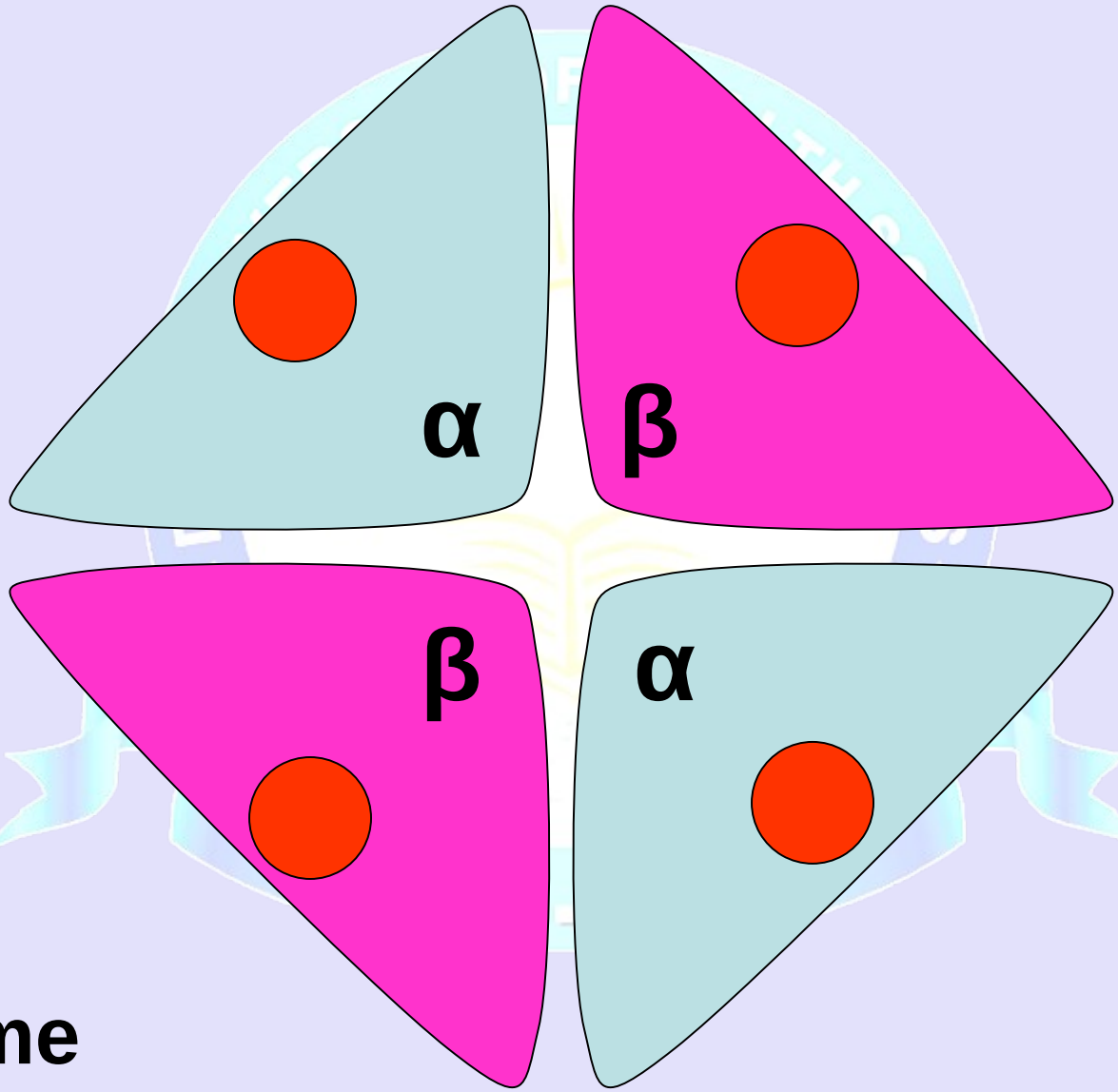


HEMOGLOBINOPATHIES

- **Hemoglobinopathies imply structural abnormalities in the globin proteins themselves**

- **Thalassemias, in contrast, usually result in underproduction of normal globin proteins, often through mutations in regulatory genes**

HEMOGLOBIN STRUCTURE



heme

Sequence of Hemoglobin changes in Human life

Embryonic Hemoglobin

- **Hgb Gower 1 (ζ_2,ε_2)**
- **Hgb Gower 2 (α_2, ε_2)**

Early embryogenesis
Yolk sac erythroblasts

Fetal Hemoglobin

- **Hgb F (α_2,γ_2)**

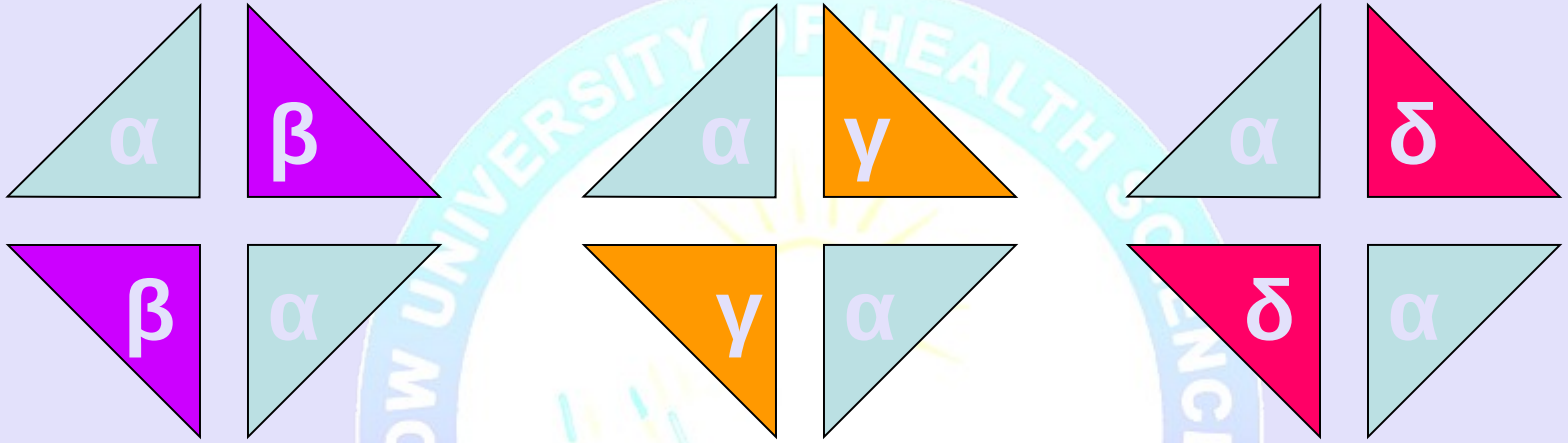
Major Hemoglobin of Intra-uterine life

Adult Hemoglobin

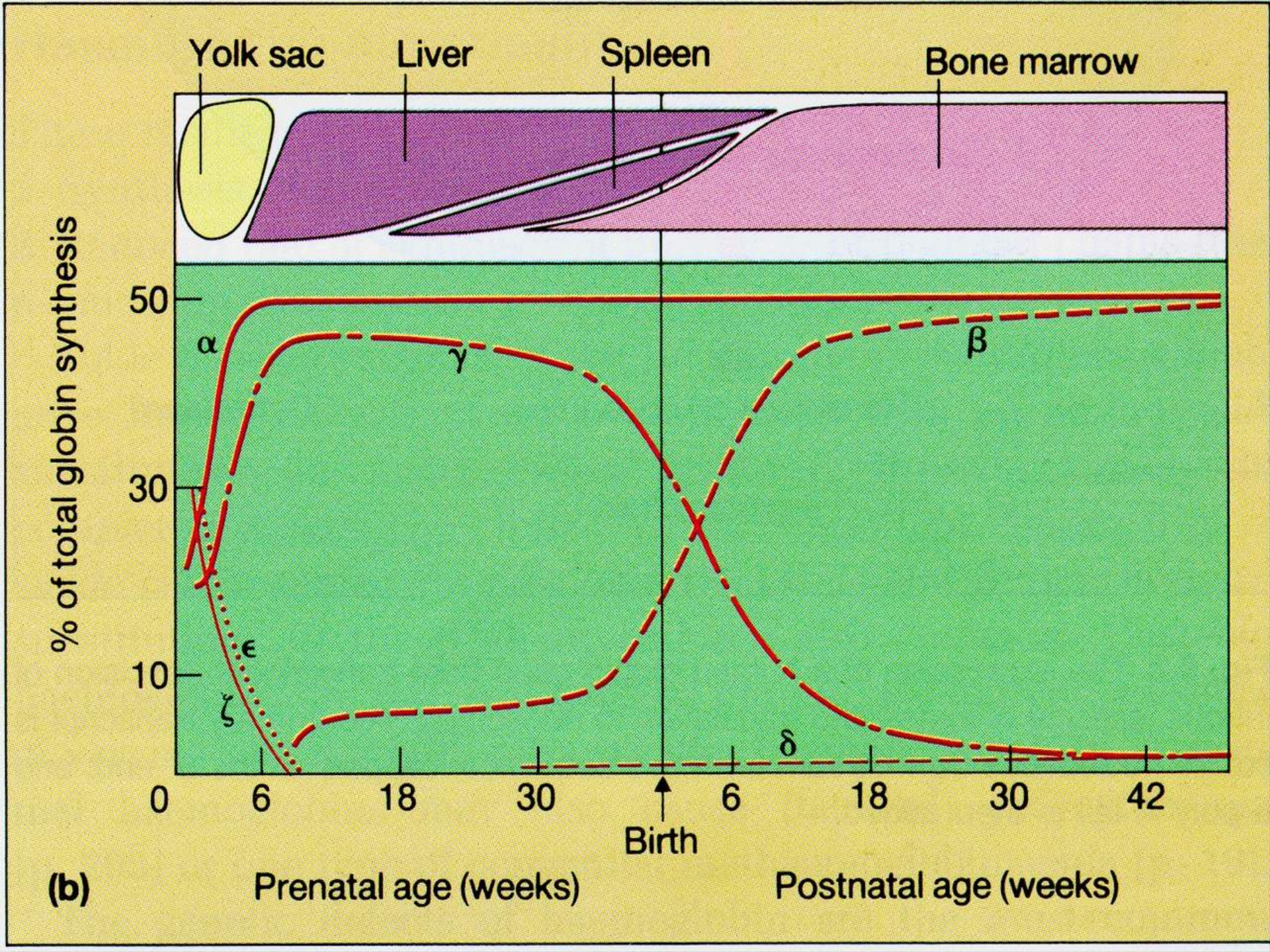
- **Hgb A (α_2,β_2)**
- **Hgb A_2 (α_2,δ_2)**
- **Hgb F (α_2,γ_2) (<1%)**

Adult Hemoglobin

HEMOGLOBINS IN NORMAL ADULTS



HbA	HbF	HbA_2
98%	~1%	<3.5%


Yolk sac
Liver
Spleen
Bone marrow
% of total globin synthesis
50
30
10
α
γ
β
ϵ
ζ
δ
0
6
18
30
Birth
6
18
30
42
(b)
Prenatal age (weeks)
Postnatal age (weeks)

HEMOGLOBINOPATHY

Examples of Hemoglobinopathy

Type of Mutation

- Single amino acid substitution
  - Hgb S, Hgb C
  - Hgb G $_{philadelphia}$
  - Hgb $_{Hammersmith}$
- Deletion
  - Hgb $_{Lepore}$
- Abnormal Hybridization
  - Hgb S, Hgb C
  - Hgb Lepore
- Abnormal elongation
  - Hgb F $_{Texas}$
  - Hgb Constant Spring

α Chain

β Chain

γ Chain

δ Chain

- Hgb $A_{2Flatbush}$

Unusual forms of Hemoglobin in the world



THE THALASSEMIAS

Syndromes in which the ***rate*** of synthesis of a globin chain is reduced

***beta* thalassemia** - reduced beta chain synthesis

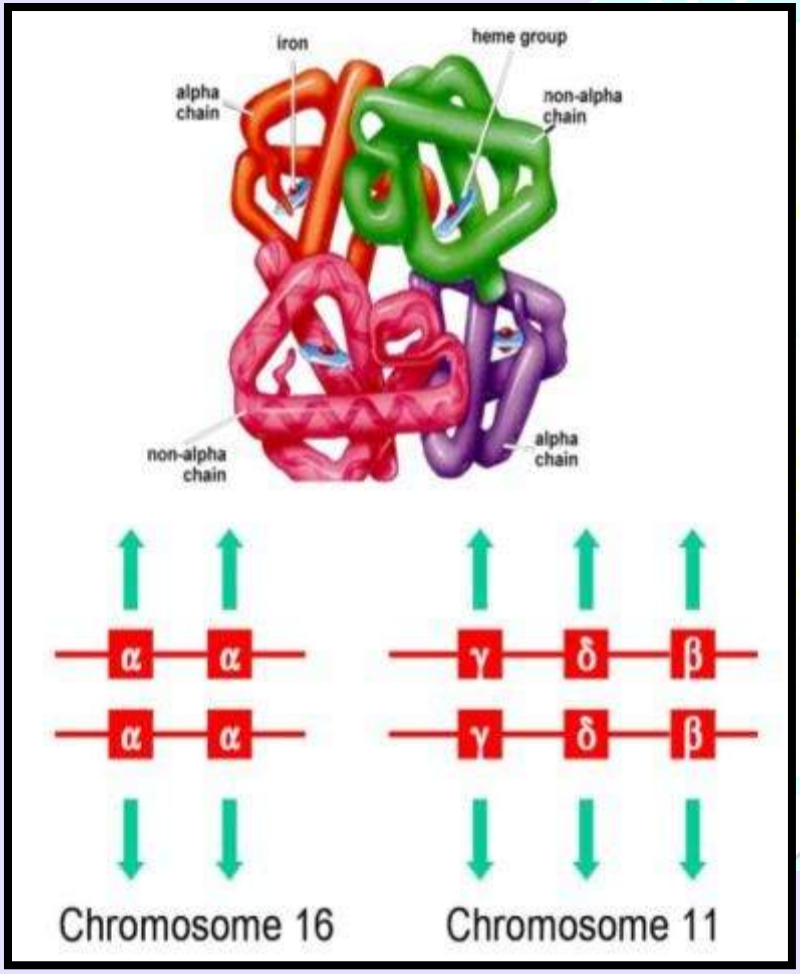
alpha *thalassemia* – reduced alpha chain synthesis

GENETICS OF THALASSEMIA

- May be either homozygous defect or heterozygous defect
- Genetic defects usually falls into one of below categories

  - Gene deletion
  - Promoter deletion
  - Nonsence mutation
  - Mutated termination
  - Splice site mutation

α AND β THALASSEMIAS



- **The α gene is duplicated: 2 genes per chromosome, therefore 4 α genes in a normal adult.**
- **The β gene is single therefore the normal adult will have two β genes**

α THALASSEMIA

Four possible α thalassemia syndromes:

1. **α-thalassemia-2 trait (silent carrier):** **One** of the four α-globin gene loci fails to function.
2. **α-thalssemia-1 trait (α Thal trait):** **Two** of the four α-globin gene loci fails to functions.
3. **Hgb H disease:** **Three** of the four α-globin gene loci fails to functions.
4. **Bart's Hydops Fetalis:** **Four** of the four α-globin gene loci fails to functions.

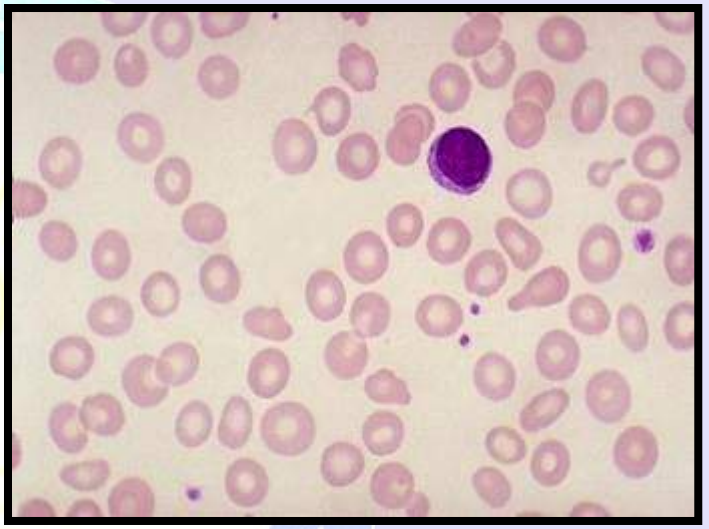
α THALASSEMIA

Syndrome	# alpha genes deleted	Newborn Hb Barts (γ4) %	Clinical Picture
Silent Carrier	1	1-2	Silent
Alpha thalassemia trait	2	3-10	Mild hypochromic, microcytic anemia
Hb H Disease	3	25	Hb H (β4) mild hemolytic anemia
Hydrops fetalis	4	80-100	Death *in utero* or shortly after birth

α THALASSEMIA

α Thalassemia Trait:

- Two of four α genes mutated
- Clinically silent
- Mild anemia
- Microcytosis

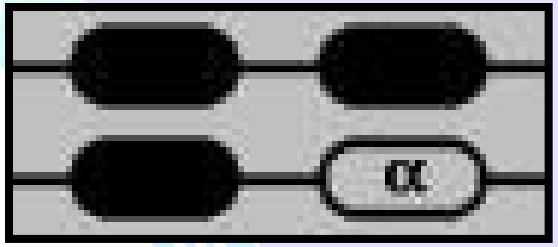


Peripheral Blood Smears

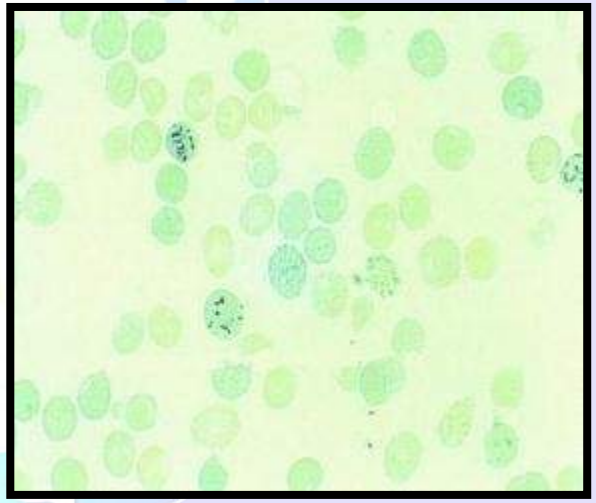


α THALASSEMIA

Hemoglobin H Disease:



- Three of four α genes mutated
- High β to α globin synthetic ratio
- Formation of Hgb H (β_4)
- Clinical features
  - Microcytic Anemia
  - Splenomegally
  - Jaundice
  - Iron overload



Peripheral Blood Smear: stained With brilliant cresyl blue, showing Precipitated Hgb H

α THALASSEMIA

Bart's Hydrops Fetalis:

- All four α genes mutated
- Presence of Barts Hgb (γ_4)
- Common in South East Asia due to the mutation
  ($--^{SEA}$), which is the cis deletion of two α genes.
- Embryonic Hgb (Hgb Gower 1 (ζ_2,ε_2)) is present but fetal Hgb (HgB F (α_2,γ_2)) is not made.
- Most babies die in utero of severe anemia and resulting hear failure



β THALASSEMIA

Types of mutations:

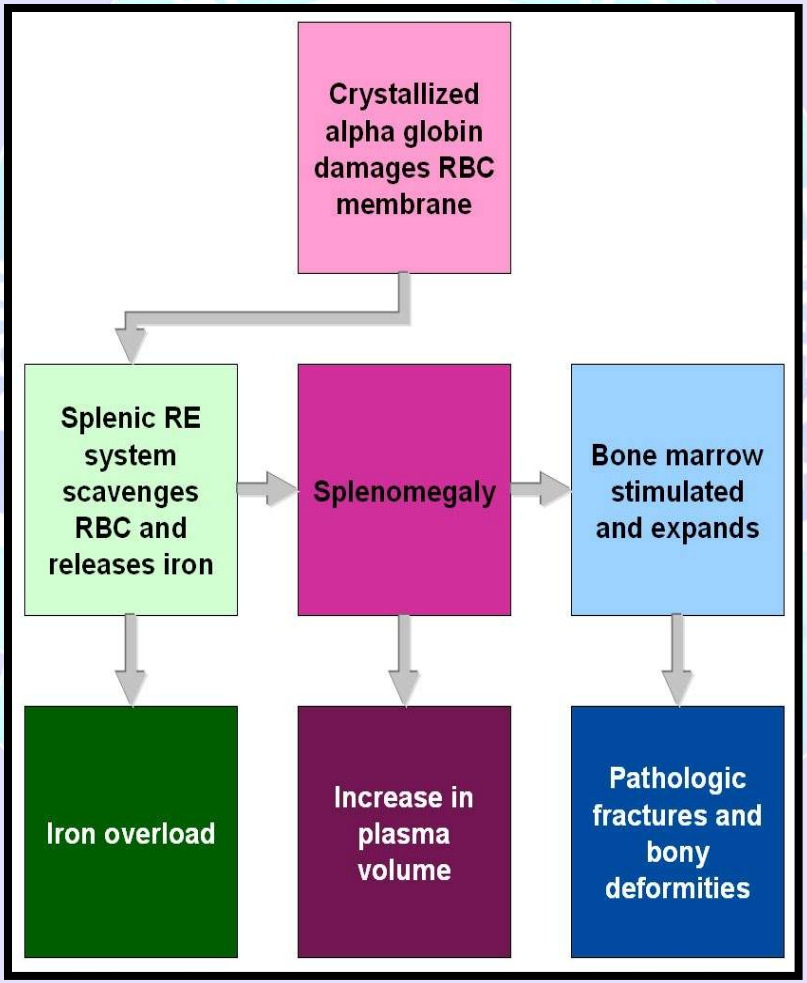
- Each chromosome 11 codes for 1 β gene, therefore the normal adult has two β genes.
- Many types of mutation can occur in the β gene:
  - Promoters and enhancers of transcription
  - Transcription
  - RNA splicing
  - Integrity of the coding sequence
  - Polyadenylation
  - Post transcriptional modification
- Different ethnic groups have different beta mutations

β THALASSEMIA MAJOR

Thalassemia Major

- Severe anemia
- Homozygous or doubly heterozygous: two seriously impaired globin genes
- Presents early in life
- Results in:
  - Retarded growth
  - Multiple organ dysfunction
  - Premature death

β THALASSEMIA MAJOR

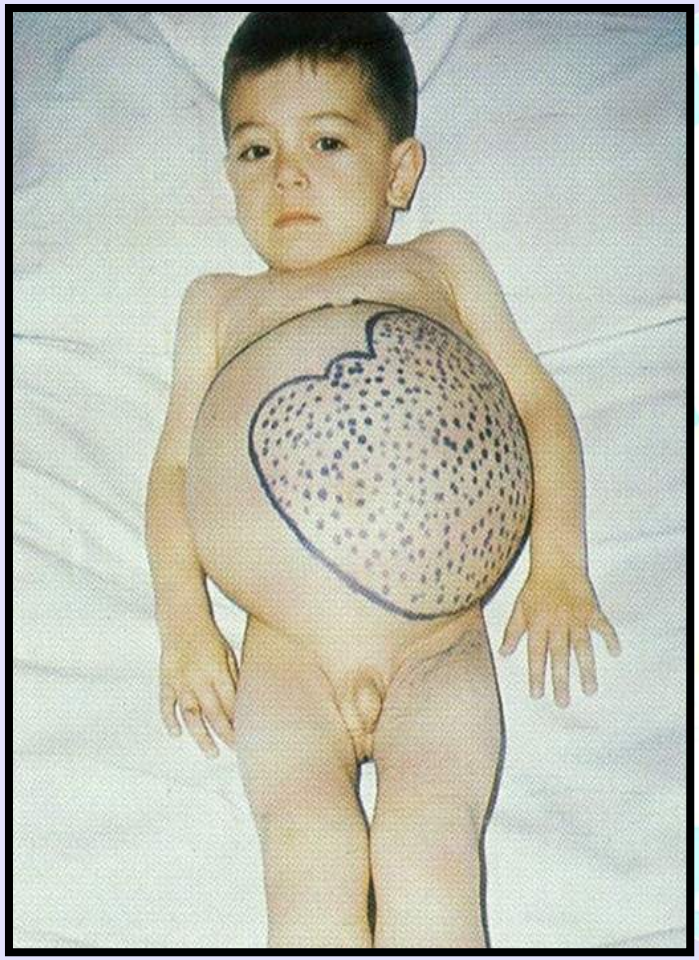


β THALASSEMIA MAJOR



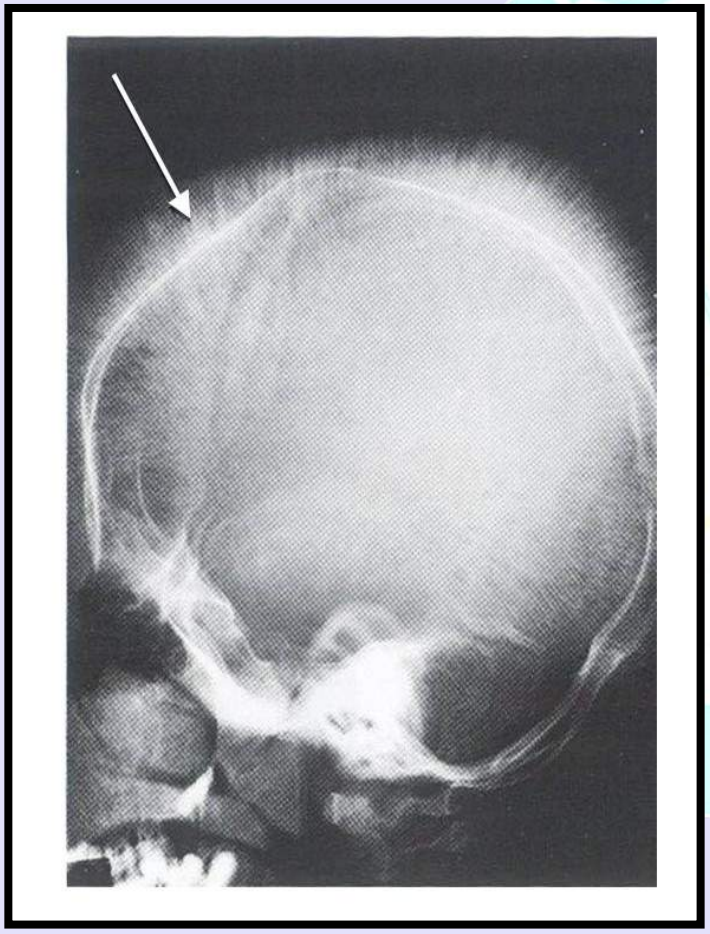
Marked bony changes from bone marrow proliferation

β THALASSEMIA MAJOR



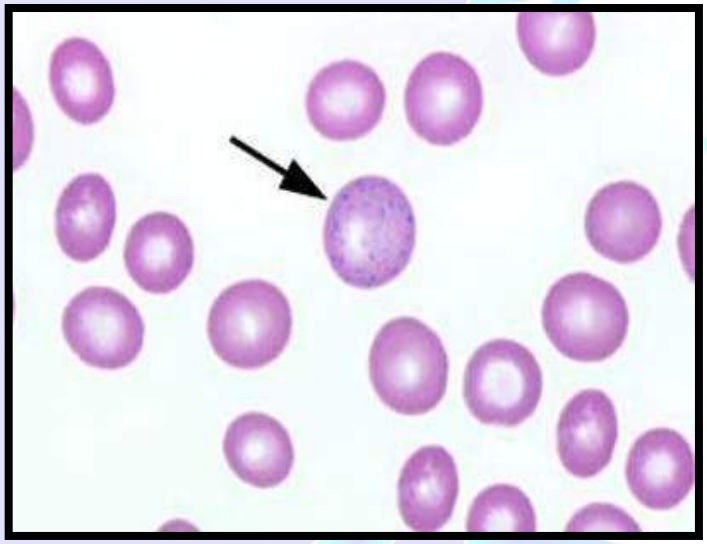
Massive splenomegally from extra-medullary hematopoiesis

β THALASSEMIA MAJOR



Bone marrow hyper expansion gives the classic “hair on end” appearance on X-ray of the skull

β THALASSEMIA MAJOR



Basophilic Stipling



Target cells and microcytosis

β THALASSEMIA INTERMEDIA

- Various genetic interactions produce the Thalassemia intermedia phenotype
  - Beta^{+} rather than beta^{0} mutation
  - Increased gamma chains available
  - Decreased alpha chains because of coinheritance of alpha deletion

- Ineffective erythropoiesis is due to excess free alpha chains that precipitate intracellularly

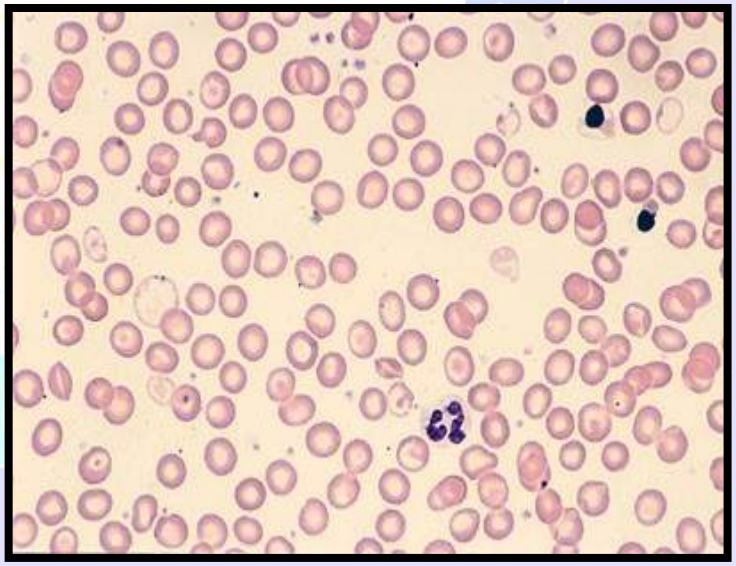
β THALASSEMIA MINOR

Thalassemia MinorTypically of little consequence clinically

- May never be diagnosed

- Minor biological defects

Mild anemia
Microcytosis



Diagnosis of Hemoglobinopathies and Thalassemia

Important Laboratory Tests:

- CBC: Anemia, microcytosis (thalassemia)
- Blood Smear: Sickle cells, target cells
- Hemoglobin Electrophoresis
- Quantitation of Hgb F, Hgb A_2, Hgb Bart's

REFFERENCES

- **HARPER'S ILLUSTRATED BIOCHEMISTRY**

  28th Edition

  Pages # 48-50 & 598

